

Topology and Manifolds

MH3600

Chapter 7: Manifolds

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Outline

- 1 Charts, Atlases, and Manifolds
- 2 Mappings between Manifolds
- 3 Some well-known two dimensional manifolds
- 4 Submanifolds
- 5 References

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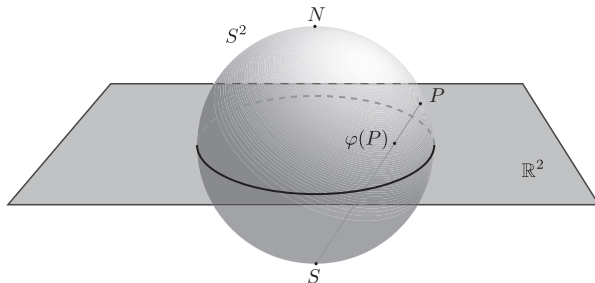
Motivation for the definition

The basic idea of a manifold is to introduce a **local object** that will support differentiation and then to **patch these local objects together smoothly**.

The sphere example.

- In $\mathbb{R}^3$ consider the **2-sphere** $\mathbb{S}^2$; that is, the set of $x \in \mathbb{R}^3$ such that $\|x\| = 1$ ($\|\cdot\|$ denotes the usual Euclidean norm).
- We can construct **bijections from subsets of $\mathbb{S}^2$ to $\mathbb{R}^2$ in several ways**. One way is to project **stereographically** from the south pole onto the plane that cuts through the equator of the sphere.
- This is a **bijection from $\mathbb{S}^2$, with the south pole removed, onto $\mathbb{R}^2$** .
- Similarly, we can interchange the roles of the poles to obtain another bijection.

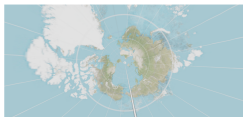
Taken from [Abraham, Marsden, Ratiu]



- With the usual subspace topology on S^2 as a subset of $\mathbb{R}^3$, each of these maps is a **homeomorphism** from their domain to $\mathbb{R}^2$.
- Each map **takes the sphere (minus the appropriate pole)** to an **open subset of $\mathbb{R}^2$** . If we go from $\mathbb{R}^2$ to the sphere by one map, then back to $\mathbb{R}^2$ by the other, we get a smooth map from an open subset of $\mathbb{R}^2$ to $\mathbb{R}^2$.

- Each map assigns a **coordinate system** to $\mathbb{S}^2$ minus a pole.
- The union of the two domains is $\mathbb{S}^2$, but **no single homeomorphism can be used between $\mathbb{S}^2$ and $\mathbb{R}^2$** ; however, **we can cover $\mathbb{S}^2$ using two of them**. In this case they are **compatible**; that is, in the region covered by both coordinate systems, **the change of coordinates is smooth**.
- For some studies of the sphere, and for other manifolds, **two coordinate systems will not suffice**. We thus allow all other coordinate systems **compatible** with these. For example, on $\mathbb{S}^2$ we want to allow spherical coordinates (θ, φ) since they are convenient for many computations.

Stereographic projections of the Earth:



$$\varphi : \mathbb{S}^2 \setminus \{S\} \rightarrow \mathbb{R}^2, \quad \varphi(x, y, z) = \left(\frac{x}{1+z}, \frac{y}{1+z} \right)$$

Mercator projection of the Earth:



$$\varphi : \mathbb{S}^2 \setminus \{N, S\} \rightarrow \mathbb{R}^2, \quad \varphi(x, y, z) = \left(\tan \frac{y}{x}, H(z) = \ln \sqrt{\frac{1+z}{1-z}} \right)$$

This chart preserves the angles (conformal) while distorting areas and distances (mapmaker's dilemma).

Charts and Atlases

Definition

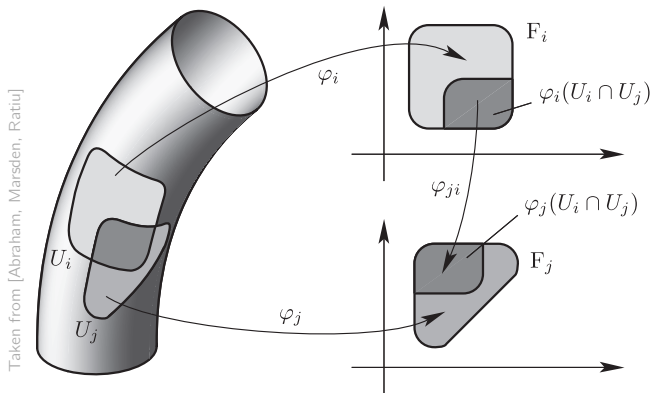
Let S be a set. A **chart** on S is a bijection φ from a subset U of S to an open subset of a Banach space. We sometimes denote φ by (U, φ) , to indicate the domain U of φ . A C^k **atlas** on S is a family of charts $\mathcal{A} = \{ (U_i, \varphi_i) \mid i \in I \}$ such that

MA1. $S = \bigcup \{ U_i \mid i \in I \}$.

MA2. Any two charts in $\mathcal{A}$ are compatible in the sense that the overlap maps between members of $\mathcal{A}$ are C^k diffeomorphisms: for two charts (U_i, φ_i) and (U_j, φ_j) with $U_i \cap U_j \neq \emptyset$, we form the **overlap map**: $\varphi_{ji} = \varphi_j \circ \varphi_i^{-1}|_{\varphi_i(U_i \cap U_j)}$, where $\varphi_i^{-1}|_{\varphi_i(U_i \cap U_j)}$ means the restriction of φ_i^{-1} to the set $\varphi_i(U_i \cap U_j)$. We require that $\varphi_i(U_i \cap U_j)$ is open and that φ_{ji} is a C^k diffeomorphism.

Note:

$f : U \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ is C^k if k times differentiable, with k -th derivative continuous
 $f : U \subset \mathbb{R}^n \rightarrow V \subset \mathbb{R}^m$ is a C^k **diffeomorphism** if C^k , invertible, and f^{-1} also C^k .



Coordinates and coordinate maps

Some terminology:

- The n -tuples in the image of a given chart (U, φ) are called the **coordinates** associated to the **coordinate system** induced by φ .
- Given $m \in M$, we say that $\varphi(m) = (x^1, \dots, x^n) \in \mathbb{R}^n$ are the **coordinates of the point** m in the given chart.
- Given $i \in \{1, \dots, n\}$ and $p_i : \mathbb{R}^n \rightarrow \mathbb{R}$, the projection onto the i th-component, we refer to the map $p_i \circ \varphi(m) = x^i$ as the **i th-coordinate map**.

Examples

Example ($\mathbb{R}^n$)

The vector space $\mathbb{R}^n$ admits an atlas formed by the [single chart](#) $(\mathbb{R}^n, \text{identity})$.

Example (The n -Sphere)

The sphere $\mathbb{S}^n$ admits an atlas formed by [two charts](#) (discussed previously for $n = 2$). Let $N = (0, 0, \dots, 0, 1)$ and $S = (0, 0, \dots, 0, -1)$ the north and south poles of $\mathbb{S}^n$. The stereographic projections from N and S are

$$\varphi_1 : \mathbb{S}^n \setminus \{N\} \rightarrow \mathbb{R}^n, \quad \varphi_1(x^1, \dots, x^{n+1}) = \left(\frac{x^1}{1 - x^{n+1}}, \dots, \frac{x^n}{1 - x^{n+1}} \right),$$

and

$$\varphi_2 : \mathbb{S}^n \setminus \{S\} \rightarrow \mathbb{R}^n, \quad \varphi_2(x^1, \dots, x^{n+1}) = \left(\frac{x^1}{1 + x^{n+1}}, \dots, \frac{x^n}{1 + x^{n+1}} \right).$$

Example (The n -Sphere (continued))

The overlap map $\varphi_2 \circ \varphi_1^{-1} : \mathbb{R}^n \setminus \{0\} \rightarrow \mathbb{R}^n \setminus \{0\}$ is given by

$$(\varphi_2 \circ \varphi_1^{-1})(z) = z/\|z\|^2, \quad z \in \mathbb{R}^n \setminus \{0\}$$

which is a C^∞ diffeomorphism of $\mathbb{R}^n \setminus \{0\}$ to itself. This formula is obtained as follows. By the definition of φ_1 , we have for $z \in \mathbb{R}^n \setminus \{0\}$

$$\varphi_1^{-1}(z) = (z^1(1 - x^{n+1}), \dots, z^n(1 - x^{n+1}), x^{n+1}) \in \mathbb{S}^n,$$

where the value x^{n+1} is fully determined by z by the constraint $\varphi_1^{-1}(z) \in \mathbb{S}^n$, that is, by the relation

$$\begin{aligned} & \| (z^1(1 - x^{n+1}), \dots, z^n(1 - x^{n+1}), x^{n+1}) \|^2 \\ &= \|z\|^2(1 - x^{n+1})^2 + (x^{n+1})^2 = 1, \end{aligned}$$

Example (The n -Sphere (continued))

and hence

$$\|z\|^2 = \frac{1 - (x^{n+1})^2}{(1 - x^{n+1})^2}.$$

Consequently,

$$\begin{aligned} (\varphi_2 \circ \varphi_1^{-1})(z) &= \left(\frac{z^1(1 - x^{n+1})}{1 + x^{n+1}}, \dots, \frac{z^n(1 - x^{n+1})}{1 + x^{n+1}} \right) \\ &= \left(\frac{z^1(1 - x^{n+1})^2}{1 - (x^{n+1})^2}, \dots, \frac{z^n(1 - x^{n+1})^2}{1 - (x^{n+1})^2} \right) \\ &= \frac{z}{\|z\|^2}. \end{aligned}$$

Definition of a Manifold

Definition

- Two C^k atlases $\mathcal{A}_1$ and $\mathcal{A}_2$ are **equivalent** if $\mathcal{A}_1 \cup \mathcal{A}_2$ is a C^k atlas.
- A C^k **differentiable structure** $\mathcal{D}$ on S is an equivalence class of atlases on S .
- The union of the atlases in $\mathcal{D}$,

$$\mathcal{A}_{\mathcal{D}} = \bigcup \{ \mathcal{A} \mid \mathcal{A} \in \mathcal{D} \},$$

is the **maximal atlas** of $\mathcal{D}$ and a chart $(U, \varphi) \in \mathcal{A}_{\mathcal{D}}$ is an **admissible local chart**. If $\mathcal{A}$ is a C^k atlas on S , the union of all atlases equivalent to $\mathcal{A}$ is called the C^k differentiable structure **generated by** $\mathcal{A}$.

- A **differentiable manifold** M is a pair $(S, \mathcal{D})$, where S is a set and $\mathcal{D}$ is a C^k differentiable structure on S .

Comments

- We shall often identify M with the underlying set S for notational convenience.
- A differentiable manifold M is a n -dimensional manifold when every chart has values in $\mathbb{R}^n$. Thus for every point $a \in M$ there is an admissible local chart (U, φ) with $a \in U$ and $\varphi(U) \subset \mathbb{R}^n$. We write $n = \dim M$.
- The definition of differentiable manifold allows the case with several connected components having different dimensions.
- If no differentiability class is explicitly given, a manifold will be assumed to be C^∞ (also referred to as “smooth”).

Example ($\mathbb{R}^n$)

The **vector space** $\mathbb{R}^n$ is a manifold. The differentiable structure is generated by the atlas that contains the single chart $(\mathbb{R}^n, \text{identity})$.

Example (The n -Sphere)

The **n -sphere** $\mathbb{S}^n$ is a manifold. The differentiable structure is generated by the atlas with the two charts described above via stereographic projections.

Example (Equivalent atlas for the sphere)

An **alternative atlas** for $\mathbb{S}^n$ has the following $2(n+1)$ charts: $(U_i^\pm, \psi_i^\pm)$, $i = 1, \dots, n+1$, where $U_i^\pm = \{x \in S^n \mid \pm x^i > 0\}$ and $\psi_i^\pm : U_i^\pm \rightarrow \{y \in \mathbb{R}^n \mid \|y\| < 1\}$ is defined by

$$\psi_i^\pm(x^1, \dots, x^{n+1}) = (x^1, \dots, x^{i-1}, x^{i+1}, \dots, x^{n+1});$$

$\psi_i^\pm$ projects the hemisphere containing the pole $(0, \dots, \pm 1, \dots, 0)$ onto the open unit ball in the tangent space to the sphere at that pole. **This atlas and the previous one are equivalent.** For example, some of the overlap maps of this atlas, for $j < i$, are given by

$$\begin{aligned} & (\psi_j^\pm \circ (\psi_i^\pm)^{-1})(y^1, \dots, y^n) \\ &= (y^1, \dots, y^{j-1}, y^{j+1}, \dots, y^{i-1}, \pm \sqrt{1 - \|y\|^2}, y^i, \dots, y^n) \end{aligned}$$

where the signs correspond. For the other overlap maps the signs in front of the square root change.

Example (Equivalent atlas for the sphere (continued))

To get a feeling for this atlas, we compute all the overlap maps for $\mathbb{S}^1$

$$\mathbb{S}^1 = \{(x^1, x^2) \in \mathbb{R}^2 \mid (x^1)^2 + (x^2)^2 = 1\}.$$

The atlas has four charts: $(U_i^\pm, \psi_i^\pm)$, $i = 1, 2$, where $U_i^\pm = \{x \in \mathbb{S}^1 \mid \pm x^i > 0\}$ and $\psi_i^\pm : U_i^\pm \rightarrow]-1, 1[$. Here is the list of all overlap maps:

$$U_1^+ \cap U_2^+ : (\psi_1^+ \circ (\psi_2^+)^{-1})(s) = \sqrt{1 - s^2}, \quad s \in]0, 1[$$

$$U_1^+ \cap U_2^- : (\psi_1^+ \circ (\psi_2^-)^{-1})(s) = -\sqrt{1 - s^2}, \quad s \in]0, 1[$$

$$U_1^- \cap U_2^+ : (\psi_1^- \circ (\psi_2^+)^{-1})(s) = \sqrt{1 - s^2}, \quad s \in]-1, 0[$$

$$U_1^- \cap U_2^- : (\psi_1^- \circ (\psi_2^-)^{-1})(s) = -\sqrt{1 - s^2}, \quad s \in]-1, 0[$$

Example (Equivalent atlas for the sphere (continued))

Here are **their inverses**

$$U_1^+ \cap U_2^+ : (\psi_2^+ \circ (\psi_1^+)^{-1})(t) = \sqrt{1 - t^2}, \quad t \in]0, 1[$$

$$U_1^+ \cap U_2^- : (\psi_2^- \circ (\psi_1^+)^{-1})(t) = \sqrt{1 - t^2}, \quad t \in]-1, 0[$$

$$U_1^- \cap U_2^+ : (\psi_2^+ \circ (\psi_1^-)^{-1})(t) = -\sqrt{1 - t^2}, \quad t \in]0, 1[$$

$$U_1^- \cap U_2^- : (\psi_2^- \circ (\psi_1^-)^{-1})(t) = -\sqrt{1 - t^2}, \quad t \in]-1, 0[.$$

Determining the domain of definition of the overlap maps is important. For example, let us check that the second and sixth maps are indeed inverses to each other: for $t \in]-1, 0[$ we have

$$\begin{aligned} ((\psi_1^+ \circ (\psi_2^-)^{-1}) \circ (\psi_2^- \circ (\psi_1^+)^{-1}))(t) &= (\psi_1^+ \circ (\psi_2^-)^{-1})(\sqrt{1 - t^2}) \\ &= -\sqrt{1 - (1 - t^2)} = -|t| = t. \end{aligned}$$

Example (The torus $\mathbb{T}^2$)

An atlas for the 2-torus can be constructed with charts of the form

$$\begin{aligned}\varphi_i : (x, y, z) \in U_i = \mathbb{T}^2 \setminus \{\text{one large circle} \cup \text{one small circle}\} \\ \longrightarrow (u_1, u_2) \in \varphi_i(U_i) =]a_i, b_i[\times]c_i, d_i[\end{aligned}$$

with u_1 longitude and u_2 latitude.

For example $\mathcal{A} = \{(U_i, \varphi_i) \mid i = 1, 2, 3\}$ with

$$\varphi_1 : \mathbb{T}^2 \setminus \{\text{exterior equator} \cup \text{one meridian}\} \rightarrow]0, 2\pi[\times]0, 2\pi[$$

$$\varphi_2 : \mathbb{T}^2 \setminus \{\text{interior equator} \cup \text{another meridian}\} \rightarrow]-\pi, \pi[\times]-\pi, \pi[$$

$$\varphi_3 : \mathbb{T}^2 \setminus \{\text{top circle} \cup \text{yet another meridian}\} \rightarrow]\pi/2, 5\pi/2[\times]\pi/2, 5\pi/2[$$

Example

A set can have more than one differentiable structure. For example, $\mathbb{R}$ has the following incompatible charts:

$$\begin{aligned}(U_1, \varphi_1) : U_1 = \mathbb{R}, \quad \varphi_1(r) &= r^3 \in \mathbb{R} \quad \text{and} \\ (U_2, \varphi_2) : U_2 = \mathbb{R}, \quad \varphi_2(r) &= r \in \mathbb{R}.\end{aligned}$$

They are not compatible since $\varphi_2 \circ \varphi_1^{-1}$ is not differentiable at the origin. Nevertheless, these two differentiable structures are “diffeomorphic” (explained later).

Differentiable structures can be “essentially different” on more complicated sets: $\mathbb{S}^7$ has two non-diffeomorphic differentiable structures is a famous result of Milnor. Similar phenomena have been found on $\mathbb{R}^4$ by Freedman and Donaldson (Exotic $\mathbb{R}^4$ - this happens only on $\mathbb{R}^4$!!).

Topology of a Manifold

Definition

Let M be a differentiable manifold. A subset $A \subset M$ is called **open** if for each $a \in A$ there is an admissible local chart (U, φ) such that $a \in U$ and $U \subset A$.

Proposition

*The open sets in M define a **topology** that has the **chart domains** as a **subbasis**. We call this the **manifold topology**.*

Proof

- The family of chart domains is clearly a subbasis.
- The family of chart domains is closed under finite intersections. Hence it is actually a basis.
- The associated topology has as open sets those determined by the following property: a subset A of M is open whenever for any $a \in A$ there exists a set $V \subset A$ that contains the element a and that can be written as a union of chart domains.
- This is equivalent to the existence of a local chart (U, φ) such that $a \in U$ and that $U \subset V \subset A$, as required. ■

Products of Manifolds

Now we show how to make the Cartesian product of two manifolds into a manifold.

Definition

Let $(S_1, \mathcal{D}_1)$ and $(S_2, \mathcal{D}_2)$ be two manifolds. The **product manifold** $(S_1 \times S_2, \mathcal{D}_1 \times \mathcal{D}_2)$ consists of the set $S_1 \times S_2$ together with the differentiable structure $\mathcal{D}_1 \times \mathcal{D}_2$ generated by the atlas $\{(U_1 \times U_2, \varphi_1 \times \varphi_2) \mid (U_i, \varphi_i) \text{ is a chart of } (S_i, \mathcal{D}_i), i = 1, 2\}$.

- The set in this definition is an atlas because given two maps $\psi_1 : U_1 \subset \mathbb{R}^n \rightarrow V_1 \subset \mathbb{R}^n$ and $\psi_2 : U_2 \subset \mathbb{R}^m \rightarrow V_2 \subset \mathbb{R}^m$, then $\psi_1 \times \psi_2$ is a diffeomorphism if and only if ψ_1 and ψ_2 are; in this case $(\psi_1 \times \psi_2)^{-1} = \psi_1^{-1} \times \psi_2^{-1}$.
- It is clear that if $\dim S_i = n_i$, then $\dim(S_1 \times S_2) = \dim S_1 + \dim S_2$.

Example

Simple examples of product manifolds are

- The **cylinder**

$$\mathbb{S}^1 \times \mathbb{R}$$

- The **n -torus**

$$\mathbb{T}^n = \mathbb{S}^1 \times \cdots \times \mathbb{S}^1 \quad (n \text{ times}).$$

- Important **4 dimensional spacetimes** are based on Cartesian products

$$\mathbb{R} \times \mathbb{S}^3, \quad \mathbb{S}^1 \times \mathbb{R}^3$$

(underlying de Sitter and anti-de Sitter spacetimes – Lorentzian analogs to the sphere and hyperboloid).

Connectedness

No assumption on the **connectedness** of a manifold has been made. In fact, in some applications the manifolds are disconnected (see tutorial). Since **manifolds are locally arcwise connected**, their **components are both open and closed**.

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Mappings between Manifolds

The following definition introduces two important ideas: the **local representative of a map** and the concept of a C^k map between manifolds. However, we are not yet ready to **differentiate these mappings**; this will be possible only after we introduce the **tangent bundle**.

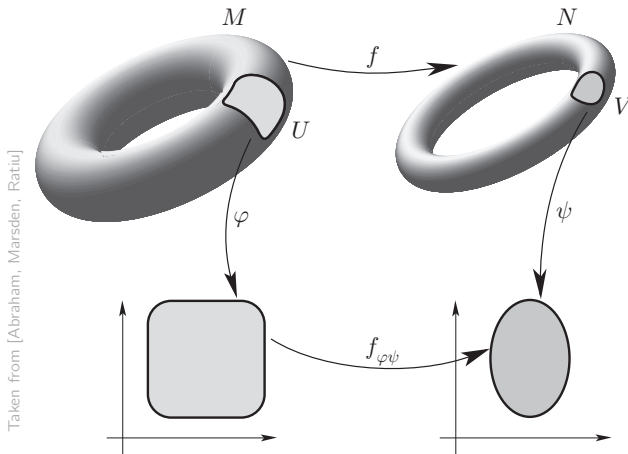
Definition

Suppose $f : M \rightarrow N$ is a mapping, where M and N are manifolds. We say f is of **class** C^k (where k is a non-negative integer), if for each x in M and admissible chart (V, ψ) of N with $f(x) \in V$, there is a chart (U, φ) of M satisfying $x \in U$, and $f(U) \subset V$, and such that the **local representative of f** ,

$$f_{\varphi\psi} = \psi \circ f \circ \varphi^{-1},$$

is of class C^k . If $k = 0$, this is consistent with the definition of continuity of f , regarded as a map between topological spaces (with the manifold topologies).

If M is an open interval in $\mathbb{R}$, f is called a C^k -**curve**.



Proposition

Let $f : M \rightarrow N$ be a continuous map of manifolds. Then f is C^k if and only if the local representatives of f relative to one collection of charts which cover M and N are C^k .

In particular, this shows that the definition of smoothness is independent of the chosen charts in maximal atlases on M and N .

Proof

Assume that the local representatives of f relative to a collection of charts covering M and N are C^k .

If (U, φ) and (U, φ') are charts in M and $(V, \psi), (V, \psi')$ are charts in N such that $f_{\varphi\psi}$ is C^k , then the composite mapping theorem and condition

MA2 show that $f_{\varphi'\psi'} = (\psi' \circ \psi^{-1}) \circ f_{\varphi\psi} \circ (\varphi' \circ \varphi^{-1})^{-1}$ is also C^k .

Moreover, if φ'' and ψ'' are restrictions of φ and ψ to open subsets of U and V , then $f_{\varphi''\psi''}$ is also C^k .

That f is C^k now follows from the fact that any chart of M can be obtained from the given collection by change of diffeomorphism, restrictions, and/or unions of domains, all three operations preserving the C^k character of f . This argument also demonstrates the converse. ■

Corollary

If $f : M \rightarrow N$ and $g : N \rightarrow P$ are C^k maps, then so is $g \circ f$.

Example

Any map from (open subsets of) $\mathbb{R}^n$ to $\mathbb{R}^m$ which is C^k in the usual sense is C^k in the sense of this definition.

Example

Other examples of C^∞ maps are the **antipodal map** $x \mapsto -x$ of S^n and the **translation map** by $(\theta_1, \dots, \theta_n)$ on $\mathbb{T}^n$ given by

$$(\exp(ir_1), \dots, \exp(ir_n)) \mapsto (\exp(i(r_1 + \theta_1)), \dots, \exp(i(r_n + \theta_n))).$$

Diffeomorphisms

Definition

A map $f : M \rightarrow N$, where M and N are manifolds, is called a C^k **diffeomorphism** if f is of class C^k , is a bijection, and $f^{-1} : N \rightarrow M$ is of class C^k . If a diffeomorphism exists between two manifolds, they are called **diffeomorphic**.

It follows from the previous proposition that the set $\text{Diff}^k(M)$ of C^k diffeomorphisms of M forms a group under composition.

Example ($\mathbb{R}$ with two manifold structures)

Take the two different manifold structures on $\mathbb{R}$: $M_1 = (\mathbb{R}, \mathcal{A}_1)$ and $M_2 = (\mathbb{R}, \mathcal{A}_2)$ with $\mathcal{A}_i = \{(U_i, \varphi_i)\}$:

$$\begin{aligned} (U_1, \varphi_1) : U_1 &= \mathbb{R}, & \varphi_1(r) &= r^3 \in \mathbb{R} & \text{ and} \\ (U_2, \varphi_2) : U_2 &= \mathbb{R}, & \varphi_2(r) &= r \in \mathbb{R}. \end{aligned}$$

Then $f : M_1 \rightarrow M_2$, $f(r) = r^3$ is a diffeomorphism since $f_{\varphi_1\varphi_2}(r) = r$.

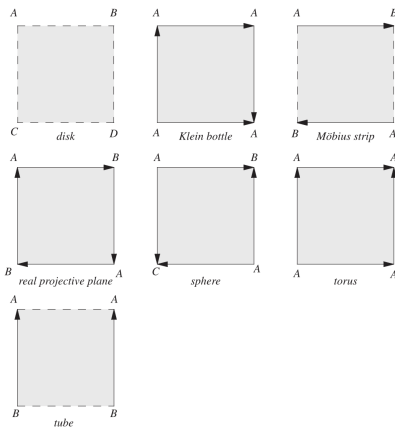
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Some well-known two dimensional manifolds

Practical approach:

plane models: underlying set S constructed from the quotient of $[0, 1] \times [0, 1]$ or $[0, 1] \times]0, 1[$ by an equivalence relation



Proposition

All these surfaces are two dimensional C^∞ manifolds.

Proof

- **Disk**: take the single chart $\varphi = id : U =]0, 1[\times]0, 1[\rightarrow]0, 1[\times]0, 1[$.
- **Möbius strip**: $(x, 0) \sim (1 - x, 1)$

$$\varphi_1 :]0, 1[\times]0, 1[\rightarrow]0, 1[\times]0, 1[, \quad \varphi_1(x, y) = (x, y)$$

$$\varphi_2 : (]0, 1[\times ([0, \epsilon[\cup]1 - \epsilon, 1])) / \sim \rightarrow]0, 1[\times] - \epsilon, \epsilon[,$$

$$\varphi_2([x, y]) = \begin{cases} (x, y) & \text{if } y \in [0, \epsilon[\\ (1 - x, y - 1) & \text{if } y \in]1 - \epsilon, 1] \end{cases}$$

Both are bijective maps, with ranges given by open subset of $\mathbb{R}^2$.

Proof.

Transition map:

$$\begin{aligned}\varphi_1 \circ \varphi_2^{-1} : \varphi_2(U_1 \cap U_2) &=]0, 1[\times (] - \epsilon, 0[\cup]0, \epsilon[) \\ &\rightarrow \varphi_1(U_1 \cap U_2) =]0, 1[\times (]0, \epsilon[\cup]1 - \epsilon, 1[)\end{aligned}$$

$$\varphi_1 \circ \varphi_2^{-1}(u_1, u_2) = \begin{cases} (u_1, u_2) & \text{if } u_2 \in]0, \epsilon[\\ (1 - u_1, u_2 + 1) & \text{if } u_2 \in] - \epsilon, 0[\end{cases}$$

is a C^∞ diffeomorphisms between open (non connected) subsets of $\mathbb{R}^2$.

- **Tube**: similar to Möbius strip
- **Klein bottle**: exercises
- **Projective plane, sphere, torus**: similar to Klein bottle.



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Submanifolds

Open submanifolds

The simplest notion of submanifold is that of open submanifold: If M is a manifold and $A \subset M$ is an open subset of M , the differentiable structure of M naturally induces one on A . We call A an **open submanifold** of M .

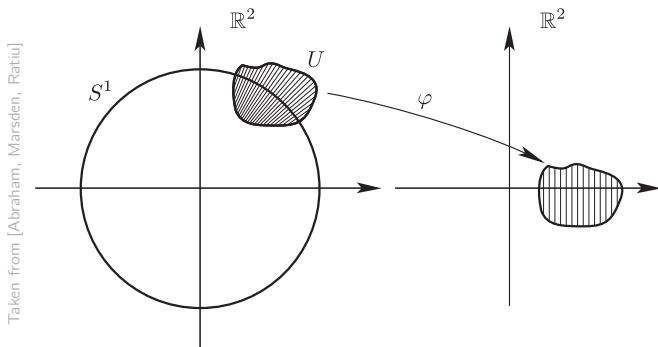
Exercise

Let M be a manifold and let (U, φ) be an **admissible chart**. Then show that

- (1) The map $\varphi : U \longrightarrow \varphi(U)$ is a **diffeomorphism** between the open manifold $U \subset M$ and the open subset $\varphi(U) \subset \mathbb{R}^n$.
- (ii) The coordinate maps $p_i \circ \varphi : U \longrightarrow \mathbb{R}$ are **differentiable**.

Hint: use the result in slide 30.

Now, we would also like to say that $\mathbb{S}^n$ is a submanifold of $\mathbb{R}^{n+1}$, although it is a closed subset. To motivate the general definition we notice that there are charts in $\mathbb{R}^{n+1}$ in which a neighborhood of $\mathbb{S}^n$ becomes part of the subspace $\mathbb{R}^n$.



From now on:

E is $\mathbb{R}^m$ for some m : it is the model space of the submanifold;

F is $\mathbb{R}^p$ for some p ;

$E \times F = \mathbb{R}^n$, $n = m + p$: the model space of the manifold.

Definition

A **submanifold** of a manifold M is a subset $S \subset M$ with the property that for each $s \in S$ there is an **admissible chart** (U, φ) of M with $s \in U$ which has the **submanifold property**, namely, that φ has the form

SM. $\varphi : U \rightarrow E \times F$ with $\varphi(U \cap S) = \varphi(U) \cap (E \times \{0\})$.

An **open subset** A of M is a submanifold in this sense. Indeed, we take $\mathbb{R} = \{0\}$ and for $x \in A$ we can use any chart (U, φ) of M for which $x \in U$ and $U \subset A$.

Proposition

Let S be a submanifold of a manifold M . Then S itself is a manifold modeled on E with differentiable structure generated by the atlas:

$$\{(U \cap S, \varphi|_{U \cap S}) \mid (U, \varphi) \text{ is an admissible chart of } M \text{ having the property } \mathbf{SM} \text{ for } S\}.$$

(Furthermore, the manifold topology on S is the subspace topology).

Proof.

- $(U \cap S, \varphi|_{U \cap S})$ are charts: $\varphi : U \rightarrow \varphi(U)$ is bijective, so its restriction $\varphi|_{U \cap S} : U \cap S \rightarrow \varphi(U \cap S)$ is also bijective. Moreover the image $\varphi(U \cap S) = \varphi(U) \cap E \times \{0\}$ is an open subset of $E \times \{0\} \simeq E$.
- **MA1**: clear $S = \bigcup \{U_i \cap S \mid i \in I\}$.
- **MA2**: given two submanifold charts $(U_i \cap S, \varphi_i|_{U_i \cap S})$ and $(U_j \cap S, \varphi_j|_{U_j \cap S})$, we have

$$\varphi_j|_{U_j \cap S} \circ (\varphi_i|_{U_i \cap S})^{-1} : \varphi_i(U_i \cap U_j \cap S) \rightarrow \varphi_j(U_i \cap U_j \cap S)$$

which can be equivalently written as

$$(\varphi_j \circ \varphi_i^{-1})|_{\varphi_i(U_i \cap U_j) \cap E \times \{0\}} : \varphi_i(U_i \cap U_j) \cap E \times \{0\} \rightarrow \varphi_j(U_i \cap U_j) \cap E \times \{0\}$$

which are C^k , bijective, inverse C^k , hence C^k diffeomorphisms.



Example ($\mathbb{S}^n \subset \mathbb{R}^{n+1}$ is a submanifold)

We show that the charts in the atlas $\{(V_i^\pm, \psi_i^\pm) \mid i = 1, \dots, n+1\}$ of $\mathbb{S}^n$ (see slide ??) come from charts $\{(U_i^\pm, \varphi_i^\pm) \mid i = 1, \dots, n+1\}$ of $\mathbb{R}^{n+1}$ with the submanifold property.

Recall:

$$\begin{aligned}\psi_i^\pm : V_i^\pm &= \{\mathbf{x} \in \mathbb{S}^n \mid \pm x^i > 0\} \rightarrow \psi_i^\pm(V_i^\pm) \\ \psi_i^\pm(x^1, \dots, x^{n+1}) &= (x^1, \dots, x^{i-1}, x^{i+1}, \dots, x^{n+1});\end{aligned}$$

with $\psi_i^\pm(V_i^\pm) = \{\mathbf{y} \in \mathbb{R}^n \mid |\mathbf{y}| < 1\}$

Consider the following charts of $\mathbb{R}^n$:

$$\begin{aligned}\varphi_i^\pm : U_i^\pm &= \{\mathbf{x} \in \mathbb{R}^{n+1} \mid \pm x^i > 0\} \rightarrow \varphi_i^\pm(U_i^\pm) \\ \varphi_i^\pm(x^1, \dots, x^{n+1}) &= (x^1, \dots, x^{i-1}, x^{i+1}, \dots, x^{n+1}, |\mathbf{x}| - 1)\end{aligned}$$

with $\varphi_i^\pm(U_i^\pm) = \{\mathbf{y} \in \mathbb{R}^{n+1} \mid y^{n+1} + 1 > \sqrt{(y^1)^2 + \dots + (y^n)^2}\}$ a cone with vertex at $(0, \dots, 0, -1)$ pointing toward the axis y^{n+1} .

Example (Continued)

It can be directly checked that

$$\varphi_i^\pm : U_i^\pm \subset \mathbb{R}^{n+1} \rightarrow \varphi_i^\pm(U_i^\pm) \subset \mathbb{R}^{n+1}$$

are C^∞ diffeomorphisms between the half spaces $U_i^\pm$ and the cones $\varphi_i^\pm(U_i^\pm)$, and form an atlas for $\mathbb{R}^{n+1}$.

In addition, they **have the submanifold property for $\mathbb{S}^n$** :

$$\varphi_i^\pm(U_i^\pm \cap \mathbb{S}^n) = \varphi_i^\pm(U_i^\pm) \cap \mathbb{R}^n \times \{0\}.$$

Therefore, **$\mathbb{S}^n$ is a submanifold of $\mathbb{R}^{n+1}$** .

Example (Graphs as submanifold)

If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a C^k map, then the graph

$$S = \{(\mathbf{x}, f(\mathbf{x})) \in \mathbb{R}^{n+1} \mid \mathbf{x} \in \mathbb{R}^n\}$$

is a C^k submanifold of $\mathbb{R}^{n+1}$.

It suffices to consider the following chart of $\mathbb{R}^{n+1}$:

$$\varphi : U = \mathbb{R}^{n+1} \rightarrow \varphi(U) = \mathbb{R}^{n+1}, \quad \varphi(\mathbf{x}, y) = (\mathbf{x}, y - f(\mathbf{x})).$$

It is C^k , bijective, with C^k inverse $\varphi^{-1}(\mathbf{x}, z) = (\mathbf{x}, z + f(\mathbf{x}))$, hence a C^k diffeomorphism.

We have

$$\begin{aligned} \varphi(U \cap S) &= \{\varphi(\mathbf{x}, y) \mid \mathbf{x} \in \mathbb{R}^n, y = f(\mathbf{x})\} \\ &= \{(\mathbf{x}, y - f(\mathbf{x})) \mid \mathbf{x} \in \mathbb{R}^n, y = f(\mathbf{x})\} \\ &= \mathbb{R}^n \times \{0\} = \varphi(U) \cap (\mathbb{R}^n \times \{0\}) \end{aligned}$$

so that (U, φ) has the submanifold property for S .

Example (Graphs of continuous maps as manifolds?)

If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a continuous (not necessarily differentiable) map, then the graph

$$S = \{(\mathbf{x}, f(\mathbf{x})) \in \mathbb{R}^{n+1} \mid \mathbf{x} \in \mathbb{R}^n\}$$

can be endowed with a manifold structure!!

Consider (U, φ) given by $U = S$ and

$$\varphi : S \rightarrow \mathbb{R}^n, \quad \varphi(\mathbf{x}, f(\mathbf{x})) = \mathbf{x}.$$

MA1: satisfied since $U = S$; **MA2:** we have $\varphi(U) = \mathbb{R}^n$ open and φ is bijective, with inverse $\varphi^{-1}(\mathbf{x}) = (\mathbf{x}, f(\mathbf{x}))$. So S is a manifold with atlas given by this single chart.

In addition S is diffeomorphic to $\mathbb{R}^n$: $\varphi : S \rightarrow \mathbb{R}^n$ is a diffeomorphism since its local expression is the identity.

However S is not a submanifold of $\mathbb{R}^{n+1}$.

For example $\{(x, |x|) \in \mathbb{R}^2 \mid x \in \mathbb{R}\}$ is a one dimensional manifold on its own, but not a submanifold of $\mathbb{R}^2$.

Proposition

Let M be a manifold of class C^k and $S \subset M$ is a submanifold. Then:

- (1) The inclusion $i_S : S \rightarrow M$ is a C^k map;
- (2) If $f : M \rightarrow N$ is a C^k map then its restriction $f|_S : S \rightarrow N$ is also a C^k -map;
- (3) A function $f : N \rightarrow S$ is of class C^k if and only if $i_S \circ f : N \rightarrow M$ is of class C^k .

Proof

- (1) The local expression of i_S with respect to a chart (U, φ) which has the submanifold property is

$$\varphi \circ i_S \circ (\varphi|_{U \cap S})^{-1} : \varphi(U \cap S) \rightarrow \varphi(U), \quad \varphi \circ i_S \circ (\varphi|_{U \cap S})^{-1}(x) = (x, 0),$$

which is clearly C^k .

- (2) Clear since $f|_S = f \circ i_S$ and by (1).

Proof.

(3) If $f : N \rightarrow S$ is of class C^k , then $i_S \circ f : N \rightarrow M$ is of class C^k by (1). Conversely, assume that $i_S \circ f : N \rightarrow M$ is of class C^k . Given $x \in N$, since $f(x) \in S \subset M$, we can choose a chart (U, φ) on M with $f(x) \in U$ with the submanifold property for S . We can also choose chart (V, ψ) on N with $x \in V$, such that $(i_S \circ f)(V) = f(V) \subset U$.

To show that $f : M \rightarrow S$ is C^k , we have to show that

$(\varphi|_{U \cap S}) \circ f \circ \psi^{-1} : \psi(V) \rightarrow \varphi(U) \cap (E \times \{0\})$ is C^k . This is true since this is the same map as $\varphi \circ (i_S \circ f) \circ \psi^{-1} : \psi(V) \rightarrow \varphi(U)$ which is C^k since f is. □

Outline

- 1 Charts, Atlases, and Manifolds
- 2 Mappings between Manifolds
- 3 Some well-known two dimensional manifolds
- 4 Submanifolds
- 5 References**

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